Harmonic Divisions and Harmonic Quadrilateral

Diana Tolu

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Definition. Let A, B, C, D be points lying on a circle. The *cross-ratio* $(A, B; C, D)$ of these four points is defined as:

$$(A, B; C, D) = \pm\frac{CA}{CB} : \frac{DA}{DB}$$

where we take the $+$ sign if the segments AB and CD do not intersect, and the $-$ sign otherwise.

Definition. Let A, B, C, D be four points on a line in this order. If $(A, C; B, D) = -1$, then $(A, C; B, D)$ is called a *harmonic division*.

Definition. Let A, B, C, D be four points lying on a circle in this order. If $(A, C; B, D) = -1$, then the quadrilateral $ABCD$ is called a *harmonic quadrilateral*.
In other words, a cyclic quadrilateral $ABCD$ is harmonic if and only if: $AB \cdot CD = DA \cdot BC$..

Properties:

1. $(A, C; B, D)$ is harmonic if and only if

$$MB \cdot MD = MA^2,$$

   where M is the midpoint of the segment AC.

2. In a triangle ABC, consider three points X, Y, Z on the interior of sides BC, CA, and AB, respectively. Let X' be the point of intersection of the line YZ with the extended side BC.

   Then $(B, C; X, X')$ is a harmonic division if and only if the cevians AX, BY, CZ are concurrent.

3. Let A, B, C' be three points lying on a circle ω. Let the tangents at A and C to ω intersect at a point P, and let the line PB intersect ω again at D. Then $ABCD$ is a harmonic quadrilateral.

   Then $(B, C; X, X')$ is a harmonic division if and only if the cevians AX, BY, CZ are concurrent.

4. Let A, B, C, D be four points lying in this order on a line d, and let P be a point not lying on this line. Take another line d' and consider the intersections A', B', C', D' of the lines PA, PB, PC, and PD, respectively, with d'. Then

$$(A, C; B, D) = (A', C'; B', D').$$

5. Let A, B, C, D be four points lying in this order on a line d. If X is a point not lying on this line, then if two of the following three propositions are true, the third is also true:

   (a) $(A, C; B, D)$ is harmonic.

   (b) XB is the internal angle bisector of $\angle AXC$.

   (c) $XB \perp XD$.

Problems:

1. **[Butterfly Theorem]** Let AB, CD, and XY be three chords of a circle ω that are all concurrent at a point M. Assume, without loss of generality, that points A and C lie on the same side of line XY. Let $R = AD \cap XY$ and $S = BC \cap XY$. Then, if M is the midpoint of XY, it is also the midpoint of RS.

2. **[BMO 2018]** Quadrilateral $ABCD$ is inscribed in circle Γ with $AB > CD$ and $AB \nparallel CD$. Let $M = AC \cap BD$. Let E be the projection of M on AB. If EM bisects $\angle CED$ prove that AB is a diameter of Γ.

3. **[TASIMO 2025]** Points A, B, C, D lie on a semicircle with diameter AB. Let $P \in AD, Q \in BC$ be such that $AP = PB$ and $CQ = QB$. Circle ω through P and Q is tangent to CD and intersects the semicircle again at R. Prove that PR, CQ and AB are concurrent.

4. **[Chinese TST]** Let $ABCD$ be a convex quadrilateral. Define the points $E = AB \cap CD$, $F = AD \cap BC$, $P = AC \cap BD$. Let O be the foot of the perpendicular from P to the line EF. Prove that $\angle BOC = \angle AOD$.

5. **[IMO 1995 Shortlist]** Let ABC be a triangle, and let D, E, F be the points of tangency of the incircle of triangle ABC with the sides BC, CA, and AB, respectively. Let X be a point in the interior of triangle ABC such that the incircle of triangle XBC touches XB, XC, and BC at Z, Y, and D, respectively. Prove that $EFZY$ is cyclic.

6. **[APMO 2013]** Let $ABCD$ be a quadrilateral inscribed in a circle ω, and let P be a point on the extension of AC such that PB and PD are tangent to ω. The tangent at C intersects PD at Q and the line AD at R. Let E be the second point of intersection between AQ and ω. Prove that the points B, E, and R are collinear.

7. **[Vietnam TST]** Two circles ω_1 and ω_2 intersect at points A and B. Let ℓ be one of their common tangents, which touches ω_1 at P and ω_2 at T. The tangents at P and T to the circle (APT) intersect at point S. Denote by H the reflection of point B across the line PT. Prove that the points A, H, and S are collinear.

8. **[ELMO Shortlist]** Let $AB = AC$ in triangle ABC, and let D be a point on segment AB. The tangent at D to the circumcircle ω of triangle BCD intersects AC at point E. The other tangent from E to ω touches it at point F. Let $G = BF \cap CD$ and $H = AG \cap BC$. Prove that $BH = 2HC$.

9. **[Vietnam TST]** Two circles ω_1 and ω_2 intersect at points A and B. The tangents to ω_1 at A and B intersect at K. Let M be an arbitrary point on ω_1 such that $M \neq A$ and $M \neq B$.

   AM intersects ω_2 at P, KM intersects ω_1 at C, and AC intersects ω_2 at Q.

   (a) Prove that the midpoint of (PQ) lies on line MC.

   (b) Prove that line PQ passes through a fixed point as M varies on the circle ω_1.

10. **[IMO Shortlist 2004 G8]** Given a cyclic quadrilateral $ABCD$, let M be the midpoint of side CD, and let N be a point on the circumcircle of triangle ABM. Assume that the point N is different from M and satisfies $\frac{AN}{BN} = \frac{AM}{BM}$.

    Prove that the points E, F, and N are collinear, where $E = AC \cap BD$ and $F = BC \cap DA$.